Introduction to Topology II: Homework 7

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Problem 1. Let $f : \mathbb{S}^n \to \mathbb{S}^n$ be a map with degree $\deg(f) \neq 1$. Show that there is a point $x \in \mathbb{S}^n$ with $f(x) = -x$.

Solution. Assume that $f(x) \neq -x$ for every $x \in \mathbb{S}^n$. Then, we can define a homotopy $H \times \mathbb{S}^n \times I \to \mathbb{S}^n$ by

$$H(x,t) = \frac{(1-t)f(x) + tx}{\|(1-t)f(x) + tx\|}.$$

Since the $f(x) \neq -x$, the denominator is never zero, so H is well-defined. At $t = 0$, we get $H(x, 0) = f(x)$, and at $t = 1$, we get $H(x, 1) = x$. Therefore, H is a homotopy between f and $\mathrm{Id}_{\mathbb{S}^n}$, which is a contradiction, since maps that are homotopic have the same degree. Thus, there must be a point $x \in \mathbb{S}^n$ with $f(x) = -x$. ☐

Problem 2. Let $f : \mathbb{S}^n \to \mathbb{S}^n$ be a map with n even. Show that there is a point $x \in \mathbb{S}^n$ with $f(f(x)) = x$.

Solution. Let $f : \mathbb{S}^{2k} \to \mathbb{S}^{2k}$ for some integer k. The only way for $f(x) \neq -x$, the $\deg f \neq -1$ (by the contrapositive of Corollary 9.3). Since $\deg f \circ f = (\deg f)^2 = 1$, then $\deg f = 1$. Therefore, $f \sim \mathrm{Id}_{\mathbb{S}^{2k}}$. Since $\mathrm{Id}_{\mathbb{S}^{2k}} \circ \mathrm{Id}_{\mathbb{S}^{2k}} = \mathrm{Id}_{\mathbb{S}^{2k}}$, then $f \circ f \sim \mathrm{Id}_{\mathbb{S}^{2k}}$. Thus, by Corollary 9.4, there exists a point $x \in \mathbb{S}^{2k}$ with $f(f(x)) = x$. ☐

Problem 3. Show that, if n is odd, $-\mathrm{Id}_{\mathbb{S}^n} \sim \mathrm{Id}_{\mathbb{S}^n}$. (Hint: Try doing it when $n = 1$ first.)

Solution. Let $f(x) = \mathrm{Id}_{\mathbb{S}^n}$ and $g(x) = -\mathrm{Id}_{\mathbb{S}^n}$. We have $\deg f = 1$. Since $g(x)$ has no fixed points, then by Corollary 9.3, we have $\deg g = (-1)^{n+1} = 1$ (since n is odd). Thus, $f \sim g$, so $-\mathrm{Id}_{\mathbb{S}^n} \sim \mathrm{Id}_{\mathbb{S}^n}$. ☐

Problem 4. Let K be a nonempty connected simplicial complex, and suppose that $f : |K| \to |K|$ is homotopic to a constant map. Show that there exists $x \in |K|$ with $f(x) = x$.

Solution. Since $f \sim g$, where $g(x) = x_0$, for every x, then $f_* = g_*$. On $H_0(K)$, we have $f_* = g_* = \mathrm{Id}_{H_0(K)}$, since K is connected. Therefore, $\Lambda_f = 1$. By the Lefschetz fixed point theorem, there exists a point $x \in |K|$ with $f(x) = x$. ☐

Problem 5. Suppose that $f : |K| \to |K|$ is simplicial and that the set $S := \{x \mid f(x) = x\}$ is finite. You may assume that every element of S is a vertex of K^1, which we stated but did not prove earlier in the course. More precisely, $S = \{\hat{A} \mid A \in K, f(A) = A\}$.

(i) Show that, if $f(A) = A$, then there is no proper face $B \subsetneq A$ such that $f(B) = B$.

(ii) Suppose that $f(A) = A$. Show that one can order the vertices of $A(v_0, \ldots, v_i)$ such that $f(v_j) = v_{j+1}$ for all $0 \leq j \leq i-1$ and $f(v_i) = v_0$.

(iii) Let $\sigma = (v_0, \ldots, v_i)$ be the orientation of A that you constructed in part (ii). Show that $f_i(\sigma) = (-1)^i\sigma$.

(iv) Show that $\Lambda_f = |S|$.

Solution to (i). Assume that there is a proper face $B \subsetneq A$ such that $f(B) = B$. Then, $\hat{B} \in S$, which contradicts the assumption that S is finite. Therefore, there is no proper face $B \subsetneq A$ such that $f(B) = B$. ☐

Solution to (ii). Let $A = (v_0, \ldots, v_i)$ be a simplex such that $f(A) = A$. Since f is simplicial, we have $f(v_j) \in A$ for all $0 \leq j \leq i$. Since there is no proper face $B \subsetneq A$ such that $f(B) = B$, then $f(v_j) \neq v_j$ for all $0 \leq j \leq i$. Therefore, we can order the vertices of A such that $f(v_j) = v_{j+1}$ for all $0 \leq j \leq i-1$ and $f(v_i) = v_0$. ☐

Solution to (iii). Let $\sigma = (v_0, \ldots, v_i)$ be the orientation of A that we constructed in part (ii). Then, we have

$$f_i(\sigma) = (f(v_0), \ldots, f(v_i)) = (v_1, \ldots, v_i, v_0) = (-1)^i\sigma.$$

Therefore, $f_i(\sigma) = (-1)^i\sigma$. ☐

Solution to (iv). Let $S = \{\hat{A}_1, \ldots, \hat{A}_m\}$, where A_j is a simplex such that $f(A_j) = A_j$ for all $1 \leq j \leq m$. Then, we have

$$\Lambda_f = \sum_{i=0}^{n} (-1)^i \operatorname{Tr}(f_i) = \sum_{j=1}^{m} (-1)^{\dim A_j} \operatorname{Tr}(f_{\dim A_j}) = \sum_{j=1}^{m} (-1)^{\dim A_j} (-1)^{\dim A_j} = m.$$

Therefore, $\Lambda_f = |S|$. $\square$